The Impact of Polysemanticity on the Locality of Neuron-level vs. Layer-level Model Editing

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1. Motivation & Background

- **The Goal: Factual Model Editing**
  - Large Language Models (LLMs) store vast amounts of factual knowledge.
  - When facts change or errors are found, it is desirable to **edit the model locally** without full retraining.
  - *Key Methods:* ROME (layer-level rank-one update), MEND, FiNE (neuron-level localization and editing).

- **The Challenge: Polysemanticity**
  - Mechanistic interpretability research suggests hidden units are often **polysemantic**: individual neurons simultaneously encode multiple unrelated features in superposition.
  - **Intuition:** If a target fact A and an unrelated fact B share many of the same underlying neurons, editing fact A might inevitably damage fact B.

2. The Central Research Question

How does polysemanticity affect the *locality* of model editing, and does operating at the *neuron level* versus the *layer level* make a difference?

Hypothesis & Approach

We hypothesize that strong mechanistic overlap between facts will fundamentally limit how local an edit can be. We test this by:

1. Building a systematic benchmark of 100 fact pairs with computed FiNE-style neuron overlap scores.
2. Defining high-overlap (SET-POLY) and low-overlap (SET-MONO) mechanistic regimes.
3. Comparing a neuron-level editor (**FiNE**) and a layer-level editor (**ROME**) on Pythia-6.9B across these regimes.